



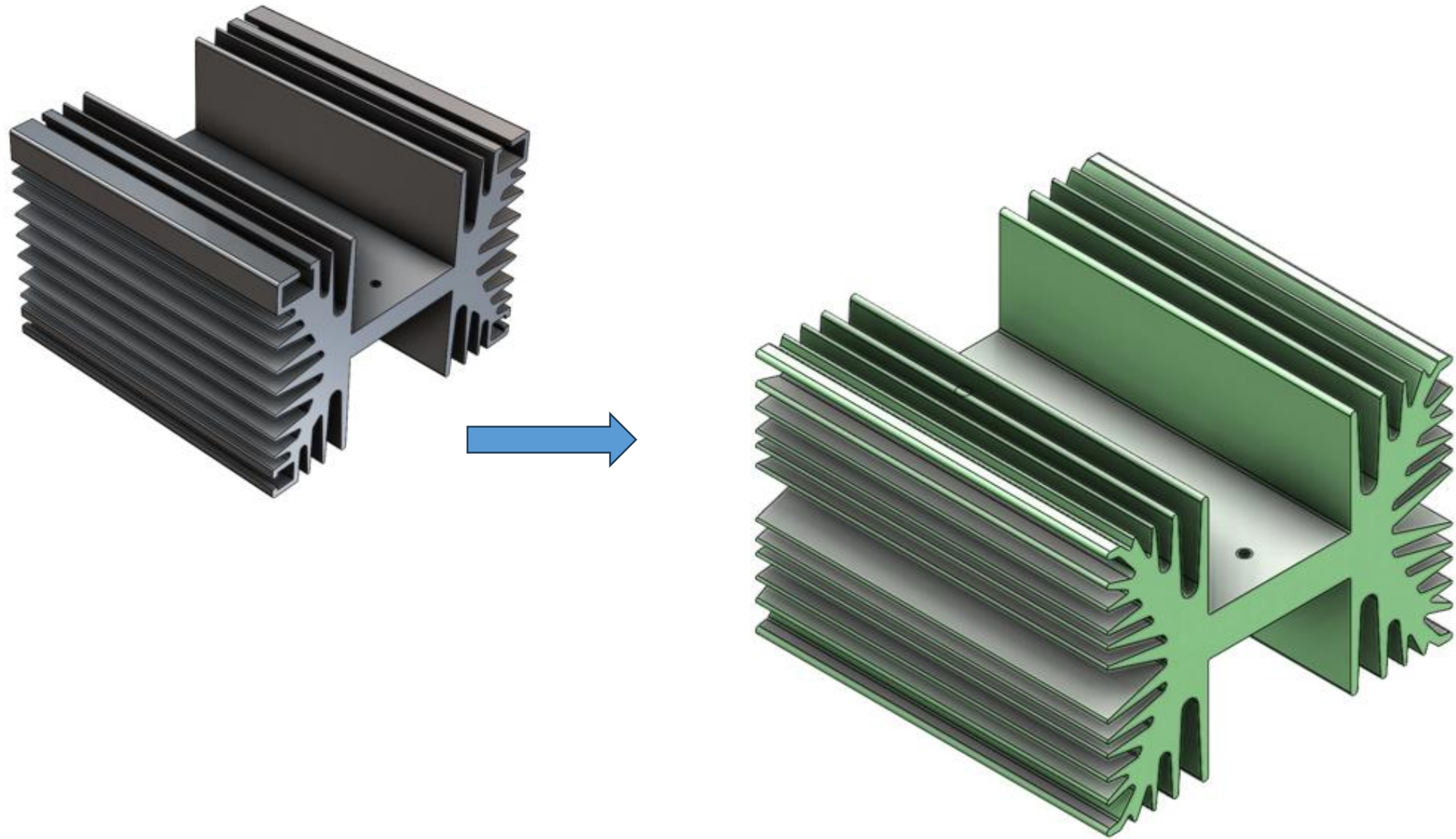
Shiretechnik Solutions Pvt Ltd

SHape Your Idea To Reality

Case Studies on Heatsink Design Optimization



CASE STUDY -1



Heatsink Profile Optimization (1/3)

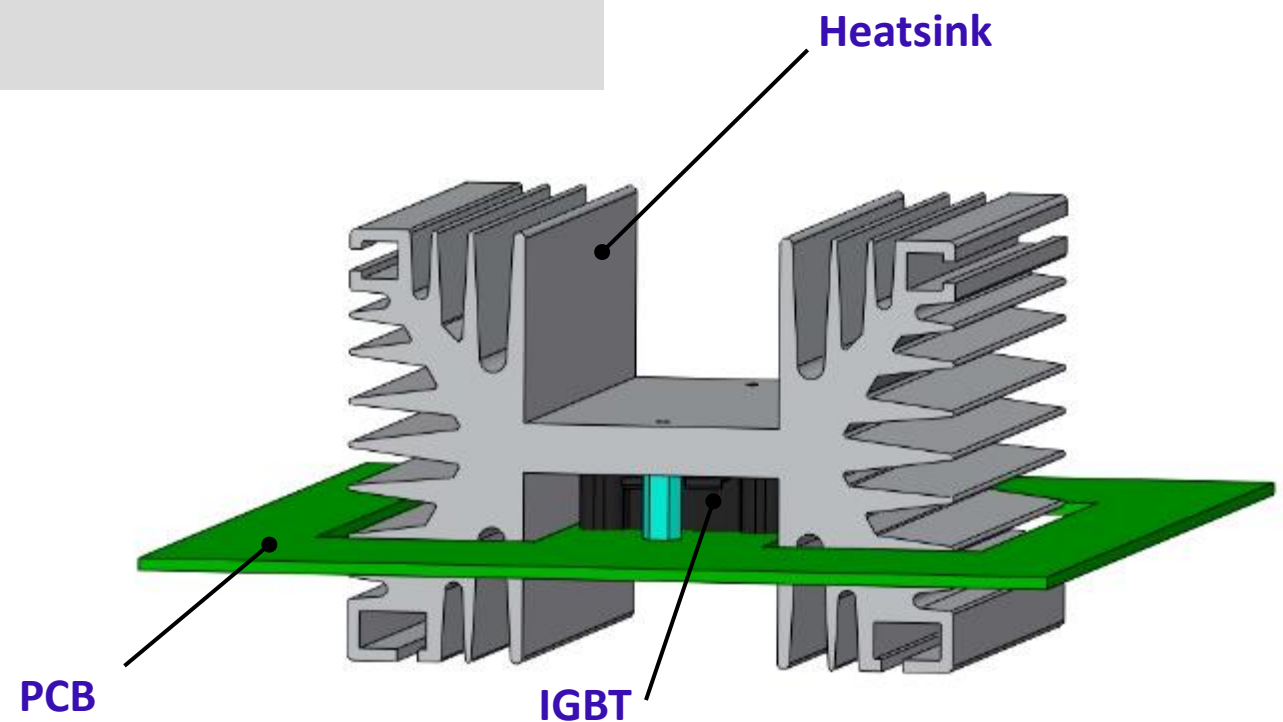
Objective of This Optimization:

Beat possible Heat Sink optimization in order to achieve

- Hot-spot temperature reduction
- Better overall thermal performance
- Compliance with DFM criteria

Thermal Design Overview:

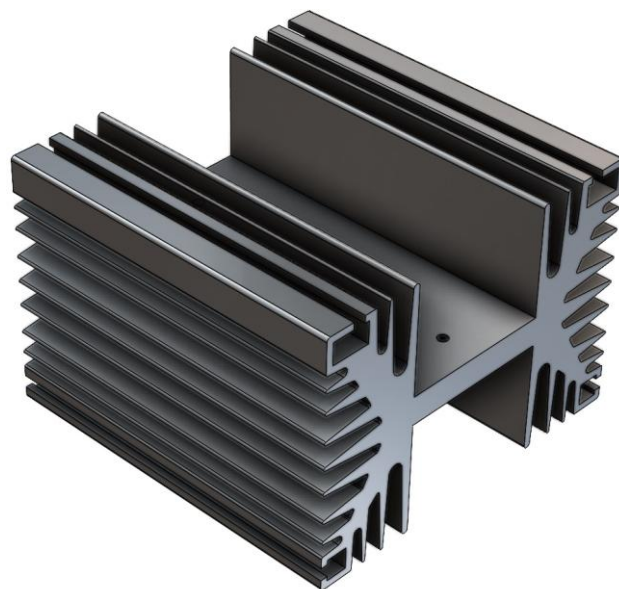
Component Type	: IGBT
IGBT's Max T _J	: 150°C
Power Dissipation	: 60 Watt
Ambient Temperature	: 45°C
Cooling Mode	: Natural Cooling
Heatsink Overall Size	: 100 x 70 x 100mm
Heatsink Material	: Al 6061
Thermal Pad Thickness	: 0.5mm
Thermal Pad Cond.	: 12 W/mK



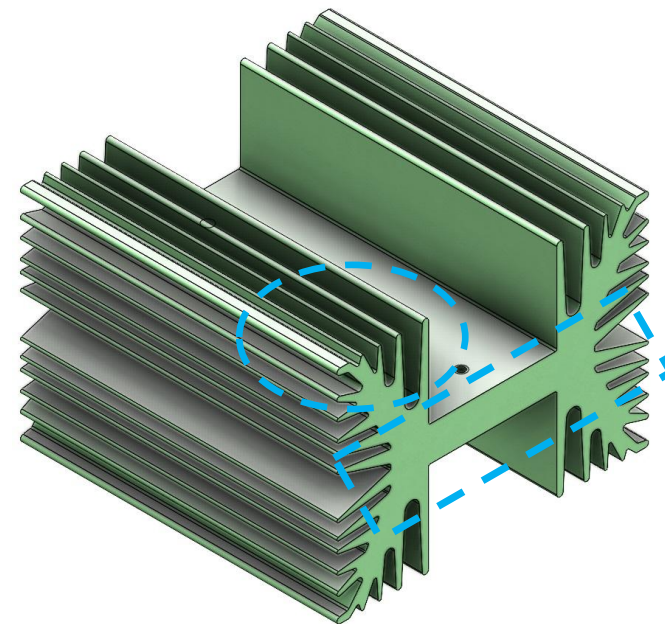
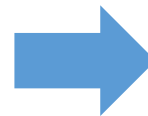
Heatsink Profile Optimization (2/3)

Optimization Considerations & Methodology:

1. Extruded Heat Sink in a suitable software simulation environment
2. Various heat sink geometrical features, etc were thoroughly investigated keeping the objective of this optimization in mind
3. Each of the scenarios was thoroughly investigated and among all the cases, a case was adjudged the best



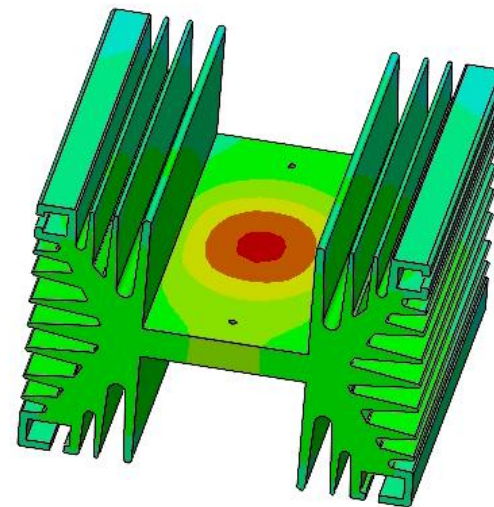
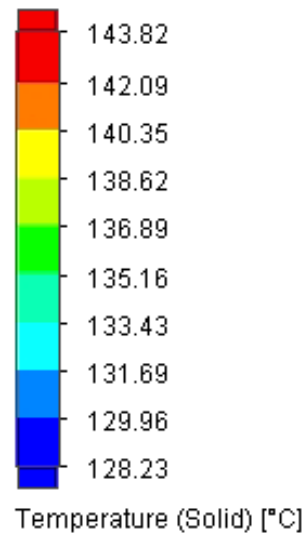
Given Heatsink Profile



Optimized Heatsink Profile

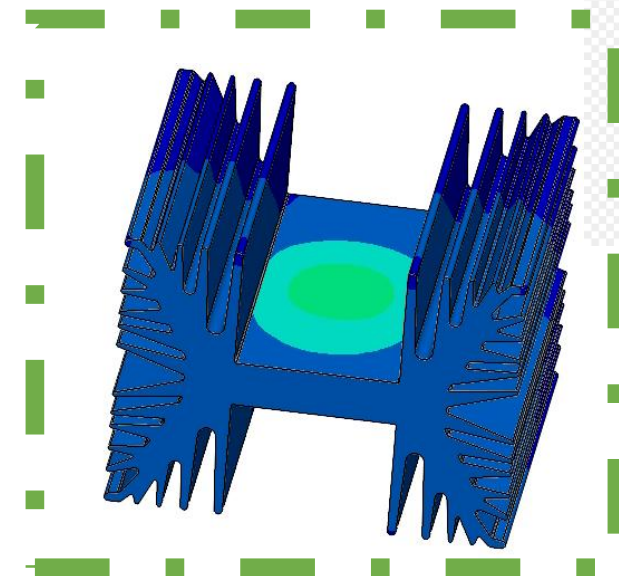
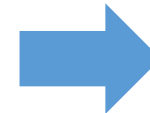
Heatsink Profile Optimization (3/3)

Optimized Case Results:



Given Heatsink

IGBT's TJ: **145.6 °C**



Optimized Heatsink

IGBT's TJ: **137.1 °C**

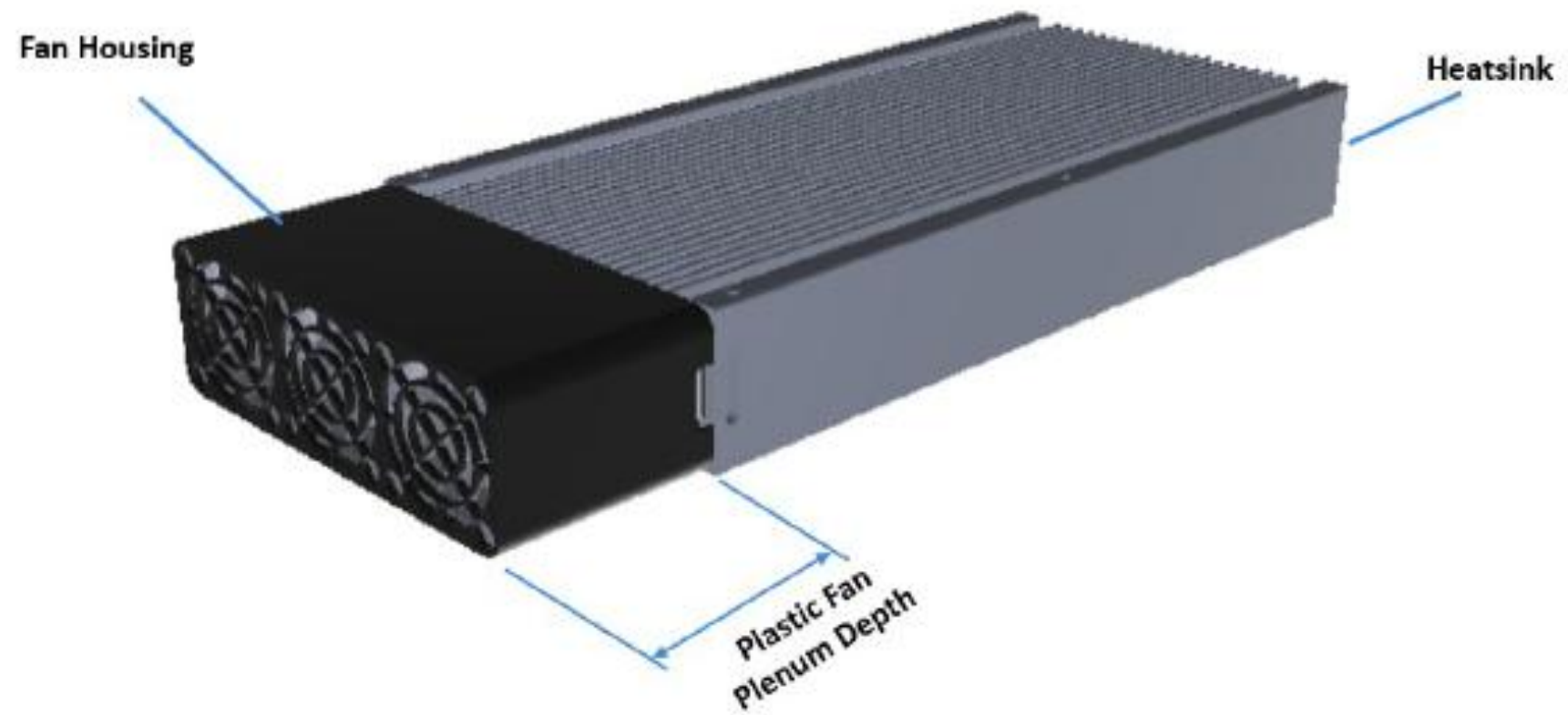


Achievements:

- ✓ ΔT achieved $> 8\text{ °C}$
- ✓ Overall cooler thermal profile
- ✓ 100% compliance with DFM considerations

(If it's within the SOW, we can also optimized to a more compact Heat Sink)

CASE STUDY -2



Heatsink Profile & Fan Plenum Optimization (1/3)

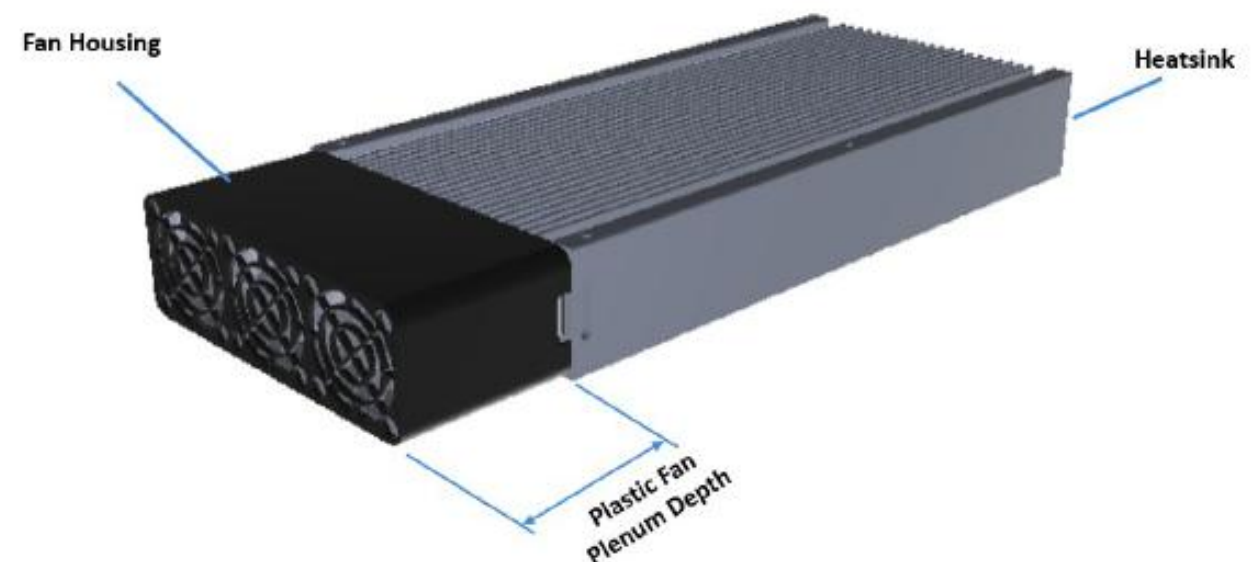
Objective of Thermal Analysis:

Beat possible Fan plenum and Heat Sink optimization in order to achieve

- Hot-spot temperature reduction
- Better overall thermal performance
- Compliance with DFM criteria

Thermal Design Overview:

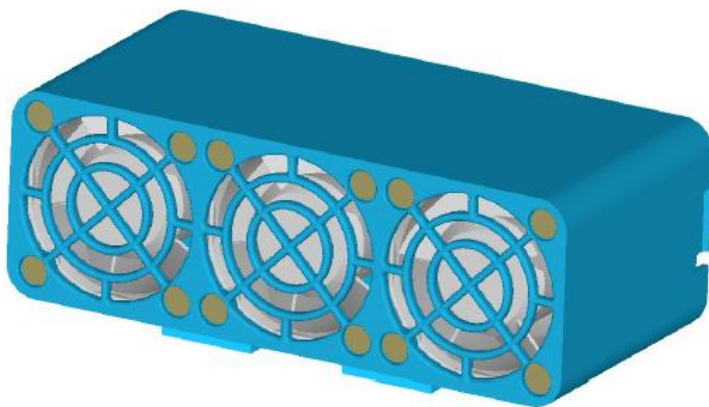
Component Type	: IGBT
Heatsink Max Temp	: 80°C
Power Dissipation	: 220 Watt
Ambient Temperature	: 55°C
Cooling Mode	: Fan-cooled
Heatsink Size	: 52 x 130 x 250 mm
Heatsink Material	: Al 6063
Grease Thickness	: 50 microns
Thermal Grease Cond.	: 0.8 W/mK
Fan Parameters	: 24 CFM, 2.2 inH2O



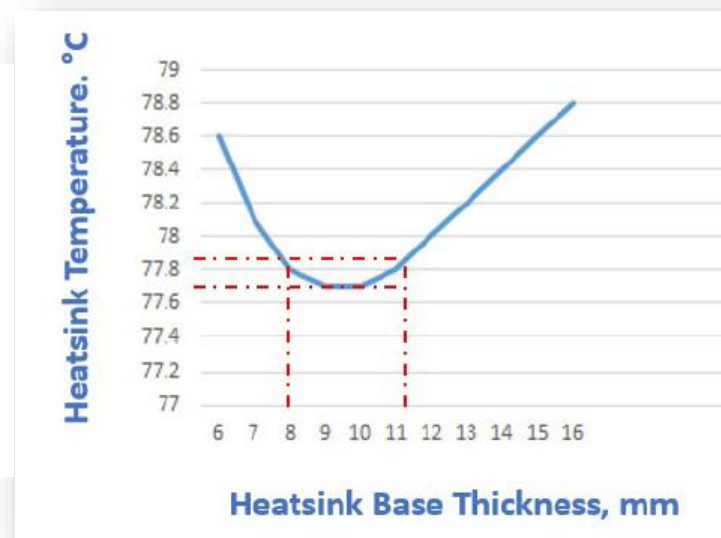
Heatsink Profile & Fan Plenum Optimization (2/3)

Optimization Considerations & Methodology:

1. Extruded Heat Sink and Fan Plenum in a suitable software simulation environment
2. A number of fans, plenum parameters, and various heat sink geometrical features, etc were thoroughly investigated keeping the objective of this optimization in mind
3. Each of the scenarios was thoroughly investigated and among all the combinations, a particular combination was adjudged the best



Fan Housing: Modified Design for Improved Performance

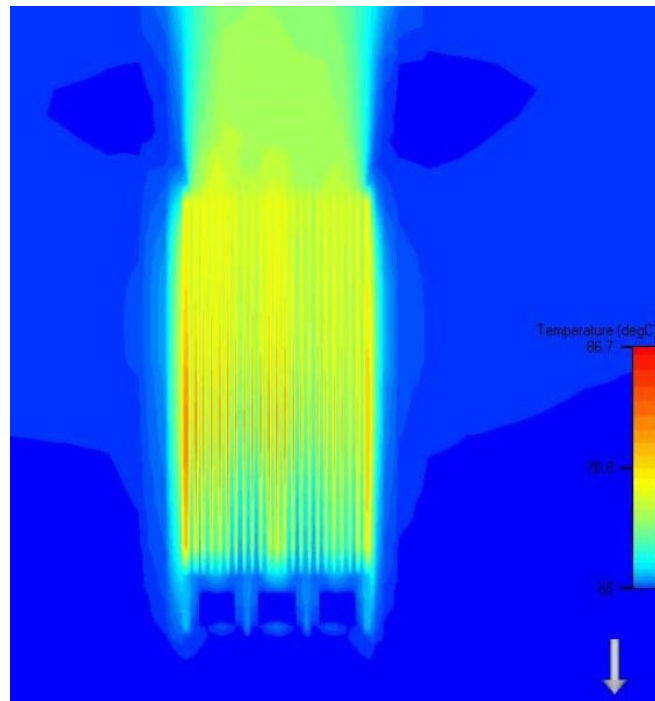


Heatsink Base Thickness Vs. Temperature



Fan Plenum Depth Vs. Static Pressure

Optimized Case Results:



Achievements:

- ✓ Heat sink maximum temperature $< 80\text{ }^{\circ}\text{C}$
- ✓ ΔT achieved = $> 7\text{ }^{\circ}\text{C}$
- ✓ Overall cooler thermal profile
- ✓ 100% compliance with DFM considerations

(Additional SOW communicated with customer for future consideration)

Original Design

Heatsink Max Temp:
84.9 °C

Optimized Design

Heatsink Max Temp:
77.7 °C



-END OF DOCUMENT-

Contact Shiretechnik Solutions

Email ID : contact@shiretechnik.com

Telephone : +91 80 23415100

**Office Address: No. 166, 5th Main, K.E.B. Layout,
Sanjaynagar, Bangalore 560 094
(Opposite to Tata Communications)**